

Mohamed Rayan Barhdadi

Curriculum Vitae · bmrayan.com

Interests

1. Physics-informed and cognitive-science approaches to vision
2. 3D/4D Scene understanding and reconstruction
3. Vision-Language Models (VLMs) and video understanding

Education

Texas A&M University Qatar 2023–2027

Bachelor of Science, Electrical Engineering

- *Thesis (3rd year): “The Physical World as the First Teacher: Structured 4D Scene Representation.”*
Advisors: Prof. Hasan Kurban, Prof. Erchin Serpedin.
- *Notable Award:*
 - Richard E. Ewing Award for Excellence in Student Research
(1 out of all students; awarded to one undergraduate annually by faculty nomination).
- *Selected Summer/Winter Schools (typically for PhD students):*
 - International Computer Vision Summer School (ICVSS), Italy, 2026.
 - African Computer Vision Summer School (ACVSS), Ghana, 2026.

Under Review

* denotes equal contribution

Intrinsic 4D Gaussian Segmentation from Scene Cues

H Yazar*, MR Barhdadi*, E Serpedin, M Tuncel, and H Kurban

British Machine Vision Conference

BMVC, 2026 (under rev.)

Perception, Not Reasoning, Limits Visual Theory of Mind

MR Barhdadi, ST Wasim, J Gall, E Serpedin, and H Kurban

Neural Information Processing Systems

NeurIPS, 2026 (under rev.)

Also Initial Version Accepted at 2nd Workshop on MMRAgI

CVPR-W, 2026

Mixture Normalizing Flows with Exact Likelihood

MR Barhdadi, SB Belhaouari, H Bensmail, and M Aupetit

Neural Information Processing Systems

NeurIPS, 2026 (under rev.)

4D Synchronized Fields: Motion-Language Gaussian Splatting

MR Barhdadi, S Abdaljalil, R Khanbayov, E Serpedin, and H Kurban

Preprint

ArXiv, 2026

Publications



IRIS: A Real-World Benchmark for Inverse Recovery and Identification of Physical Dynamic Systems from Monocular Video

R Khanbayov*, MR Barhdadi*, E Serpedin, and H Kurban

European Conference on Computer Vision

ECCV, 2026

EMPATHIA: Multi-Faceted Human-AI Collaboration for Refugee Integration

MR Barhdadi, M Tuncel, E Serpedin, and H Kurban

Neural Information Processing Systems

NeurIPS, 2025

+ Invited Keynote Talk at NeurIPS Affinity Workshop 2025

PhysicsNeRF: Physics-Guided 3D Reconstruction from Sparse Views

MR Barhdadi, H Kurban, and H Alnuweiri

Building Physically Plausible World Models Workshop

International Conference on Machine Learning

ICML-W, 2025

Research Experience	Harvard University, AI and Robotics Lab	03/2026–Present
	Research Assistant with Dr. Mengyu Wang Topic: 3D/4D Understanding, Vision Language Models.	
	Texas A&M Qatar, Kurban Intelligence Lab	02/2025–Present
	Research Assistant with Dr. Hasan Kurban and Dr. Erchin Serpedin Topics: All things Perception/Vision from Cog Sci/Phys approach.	
	Qatar Computing Research Institute, AI Center	05/2025–03/2026
	Research Intern with Dr. Halima Bensmail and Dr. Michael Aupetit Topics: Normalizing Flows and Generative Models.	
	Texas A&M Qatar, Power System Modeling Laboratory	01/2024–12/2024
	Research Assistant with Dr. Selma Awadallah Topic: ML for fault detection in electrical transformers.	
Industry Experience	Overshoot (Y Combinator W26)	05/2026–Present
	Research Scientist Intern with Younes and Zakaria El Hjouji Topic: Unified Encoder for Vision Language Models.	
	Ottonomi AI	07/2025–10/2025
	Computer Vision Software Engineer with Dr. Hussein Alnuweiri Under NDA ;)	
	Digital & Integration, SLB	06/2025–08/2025
	Engineering Intern with Azhar Khan Topic: Automated Well Integrity Evaluation.	
	ValuesLab Award Grant, Columbia University	2026
Grants and Funding	Principal Investigator, Awarded \$3,000 of funding and compute. Topic: Machine Imagination as a new learning paradigm.	
	Student Research Experience Grants (SREG), TAMUQ	2026
	Lead Undergraduate Investigator, Awarded \$50,000 of funding. Topic: Physics informed 3D reconstruction and understanding.	
	Undergraduate Research Experience Project (UREP), QORDI	2025
	Member, Awarded \$26,000 of funding. Topic: ML for fault detection in electrical transformers.	
	Travel Award to ICVSS in Italy, TAMUQ	2026
	Travel Award to CVPR in Denver, Overshoot (YC W26) and TAMUQ	2026
	Travel Award to ACVSS in Ghana, Google and INRIA	2026
	Travel Award to EACL in Morocco, TAMUQ	2026
	Travel Award to MenaML in Saudi, KAUST and Google DeepMind	2026
	Complementary Registration NeurIPS, NeurIPS Committee	2025
	Travel Award to NeurIPS in San Diego, TAMUQ	2025
	Travel Award to ICML in Vancouver, TAMUQ	2025
	Travel Funding to present at IFTP in Texas, TAMU	2025
	Travel Award for SLEP leadership exchange in Texas, TAMU	2025
	Selected Talks	
	Invited Keynote Speaker at the “MusIML” Workshop, NeurIPS 2025	12/2025
	Invited Speaker at “Human Inspired Computer Vision” Seminar, TAMUQ	11/2025

	Selected Speaker at the “Undergraduate Research Conference”, MIT	10/2025
	Invited Speaker at Moroccans in Artificial Intelligence Research (MAIR)	10/2025
Honors and Awards	Richard E. Ewing Award for Excellence in Research (1 out of 450) – \$1,250	2026
	Undergraduate Research Scholars, College of Engineering, TAMU	2026
	1st Place at Qatar Computing Research Institute Symposium (1/108)	2025
	First Place at AIX Hackathon, Rising Ventures (+85 Teams) – \$5,000	2025
	1st Place at EC 3rd Undergraduate Research Retreat – \$550	2025
	Woqod x Qatar Foundation Scholar Award Recipient – \$10,000	2025
	Second Place (Global Phase) Invent for the Planet by TAMU – \$2,500	2025
	First Place (Qatar Phase) Invent for the Planet by TAMUQ – \$1,650	2024
	+Best Prototype Award	
	+Best Video Award	
	Winner of Qatar Foundation Tech Pitch Competition – \$11,000	2024
	2nd Place Texas A&M University Qatar Robotics Competition	2024
	Inducted in Engineering Honors Program at TAMUQ	2023
Academic Service	Reviewer, NeurIPS, CreativeAI Track	2026
	Reviewer, ICML, AI4Good Workshop	2026
	Reviewer, CVPR, ReLearn: Rediscovering Intelligence Workshop	2026
	Volunteer, ICML/NeurIPS, MusIML Workshop	2025
Leadership	IEEE, <i>Student Member</i>	2023–Present
	ACL, <i>Student Member</i>	2026–Present
	The Peace Club TAMUQ, <i>Advisor</i>	2025
	The Peace Club TAMUQ, <i>President</i>	2025
	The Peace Club TAMUQ, <i>Vice-President</i>	2024
	Lead Organizer, “Effective Humanitarian Engineering Solutions Workshop”	2024
	Qatar Foundation, <i>Student Orientation Leader</i>	2024
	IEEE Student Chapter, <i>Class Representative</i>	2023
	Moroccan National Swimming Federation, <i>Instructor Volunteer</i>	2022
Languages	English (Fluent) · Arabic (Native) · French (Bilingual Proficiency) · Spanish (Elementary)	
References	Dr. Hasan Kurban hkurban@hbku.edu.qa <i>Director, Kurban Intelligence Lab</i> <i>Assistant Professor, Hamad Bin Khalifa University</i> <i>Adjunct Assistant Professor, Texas A&M University Qatar and IU Bloomington</i>	
	Dr. Erchin Serpedin eserpedin@tamu.edu <i>Department Chair, ECEN, Texas A&M University Qatar</i> <i>Full Professor, Texas A&M University</i>	
	Dr. Halima Bensmail hbensmail@hbku.edu.qa <i>Principal Scientist, Qatar Computing Research Institute</i> <i>Previously: Oak Ridge National Lab, University of Washington, INRIA</i>	
	Zakaria El Hajouji zakaria@overshoot.ai <i>Founder & CEO, Overshoot (Y Combinator W26)</i> <i>Previously: MIT, Meta, Uber</i>	